

Proportional Analysis and Microstructural Characterization of Threading Dies Manufactured Using Conventional and EDM Machining Methods [†]

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Abstract: This experimental study investigates the effects of conventional and non-conventional (Electrical Discharge Machining, EDM) techniques on the performance of threading dies fabricated from H13 die steel. The research focuses on variations in material properties, operational efficiency, and failure mechanisms. Results indicate that conventional machining significantly increases die hardness, leading to higher mechanical stress and premature failure. In contrast, EDM maintains structural integrity with minimal hardness variation. Microstructural analysis confirms superior grain structure stability in EDM-processed dies. Performance testing showed that conventional threading dies failed after threading 16 inches of mild steel rods, while EDM-fabricated dies successfully threaded up to 20 inches. These findings underscore the detrimental effect of mechanically induced stress in conventional techniques and highlight the superior durability and dimensional precision achievable through non-conventional machining methods.

Keywords: Conventional machine; mechanical stress; Microstructural analysis; Electrical Discharge Machining (EDM); grain structure; failure mechanisms.

1. Introduction

The manufacturing performance and operational lifespan of dies within the die fabrication field depend heavily on the selection of appropriate machining techniques for these critical manufacturing components which include molding tools along with threading and Taping technologies [1]. Tapping and threading dies represent essential categories which produce accurate threaded elements that serve multiple engineering applications. Material properties of these dies change substantially during machine due to variations in hardness combined with microstructure variations and the formation of residual stresses since these factors act as primary indicators of die operational life and wear capability. Traditional turning and milling operations and grinding procedures generate major residual stress because these manufacturing methods require contact between the tool and workpiece surface. The necessary shaping stress increases material hardness while making it more prone to cracking with threading operations that induce large amounts of material stress and interface friction [2]. Non-conventional machine methods including Electrical Discharge Machining (EDM) provide an alternative processing method through the avoidance of direct tool-workpiece interaction. Electrical power through electrical

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discharges to eliminate material thus diminishing residual stress formation and maintaining the die's structural condition. Electrical Discharge Machining has gained wider recognition because it produces exact high-quality dies that exhibit superior properties than traditional production methods. Threaded operations require die tools to maintain their structural strength to perform cutting operations at full capacity while avoiding failures and minimizing tool wear [3]. EDM could be a broadly utilized instrument within industry to guarantee great work on all sorts of items. Exactly machining of electronic materials notwithstanding of the hardness of steel, metal amalgams, graphite and indeed a few ceramic materials. Micro-EDM is broadly utilized within the micro-mold plan industry to make molds, cavities, and indeed complex 3D models. The primary advantage of electro discharge machining or micro-EDM over other machining forms is that it is an electro warm preparation that expels metal in any case of its hardness, and the strength and property of the workpiece are irrelevant. In this manner, the mistake caused by the misshaping constraint of the gadget is nearly zero [4]. In expansion, since there's no coordinate contact between the cathode and the workpiece, there are no splitting, stretch and vibration issues amid machining [5].

2. Literature Review

Die manufacturing plays a crucial role in various industries, particularly in metal forming and injection molding. Traditional (conventional) machining methods, such as turning, milling, and grinding, have long been utilized for die production. However, these methods encounter significant challenges due to increased material hardness, reduction in grain structure diameter, and excessively high cutting temperatures [6]. These challenges lead to undesirable alterations in material properties, thereby affecting the performance and durability of dies. Non-conventional machining (NCM) methods, such as Electrical Discharge Machining (EDM), Laser Beam Machining (LBM), and Abrasive Water Jet (AWJ), offer viable alternatives to overcome these limitations. This literature review aims to explore recent research addressing the effectiveness of non-conventional machining methods in die manufacturing and their advantages over conventional approaches [7].

2.1 Challenges in Conventional Machining of Dies

This article [8] provided a comprehensive analysis of the constraints associated with conventional machining of hard materials used in die manufacturing. The study highlighted key issues such as excessive heat generation, tool wear, and microstructural alterations, which negatively impact die performance. Conventional machining struggles with maintaining precision and surface integrity when working with hardened materials, leading to frequent tool replacement and increased manufacturing costs. Furthermore, machining-induced residual stresses often compromise the fatigue strength of dies, reducing their lifespan [9]. This article [10] extended the discussion by evaluating the microstructural effects of machining on materials. Their study demonstrated that high-temperature conditions in conventional machining result in grain structure refinement and hardness increment, leading to brittle behaviour in dies. The reduction in formability due to machining-induced structural changes underscores the need for alternative manufacturing approaches.

2.2 Advancements in Non-Conventional Machining Methods

recent advancements in non-conventional machining offer promising solutions. A novel bimetal-gradient-layer surfacing technique [11] demonstrated enhanced microstructural integrity by minimizing thermal effects and maintaining original material properties. Further research [12] into methods like Abrasive Water Jet (AWJ) and Electrical Discharge Machining (EDM) showed improved dimensional accuracy and surface finish

without damaging the microstructure. A comprehensive review [13] confirmed that EDM and Laser Beam Machining (LBM) deliver superior surface quality and extended tool life, effectively overcoming the thermal and structural limitations of conventional methods and paving the way for more efficient and durable die manufacturing.

2.3 Conventional vs. Non-Conventional Machining

A comparative analysis of conventional and non-conventional machine methods highlights the clear advantages of non-conventional approaches. While conventional machines remain prevalent for their cost-effectiveness and accessibility, they struggle with hardened materials, often leading to tool wear, poor surface quality, and undesirable microstructural changes. In contrast, non-conventional machining techniques offer higher precision, better surface finishes, and minimal thermal damage, making them more suitable for complex die manufacturing [14]. The reviewed literature underscores how these advanced methods address key challenges of traditional machining such as hardness-related difficulties and grain structure alterations—ultimately enhancing die performance and durability. Future research should aim to optimize non-conventional parameters and explore hybrid methods to further boost efficiency in die production [15]. However, there is a research gap in identifying machining methods that can overcome these limitations. Non-conventional machining procedures address this gap by offering effective machining techniques that minimize hardness, enhance generation rates, improve surface finish, extend tool life, and achieve higher precision without compromising the material's microstructure.

3. Experimental Setup

This experimental section describes the detail regarding the material composition, machining parameters for both processes, Experimental steps, and the output parameters as shown in Table 1:

Table 1: EDM Process Parameters

Parameter (Value)	Parameter Description	Rationale
High Tension (0)	Peak voltage during EDM discharge.	Set to zero to prevent arcing and dielectric failure.
Low Tension (6)	Base voltage for spark stability.	Selected from literature; refined for stable discharge [2].
Pulse On Time (0)	Duration of current flow per spark.	Minimized due to the use of Quick Discharge mode.
Pulse Off Time (200)	Cooling interval between discharges.	Literature-based; avoids overheating and maintains stability [4].
Quick Discharge On (7)	Controls rapid discharge efficiency.	Optimized experimentally for improved MRR.
Quick Discharge Up (6)	Controls electrode retraction speed.	Based on EDM studies for flushing and gap control.
Spindle Speed (3)	Electrode rotation rate.	Low RPM ensures spark uniformity and reduces wear.
SERVO Voltage (80 V)	Controls electrode–workpiece gap.	Tuned for consistent discharge initiation.
Quick Approach Voltage (50 V)	Voltage for rapid tool positioning.	Adjusted for setup speed and minimal tool wear.
Polarity (+)	Electrode polarity direction.	Positive polarity is preferred for steel in EDM.
MRR (480 mm ³ /min)	Volume of material removed per unit time.	Trial-validated; aligns with H13 machining benchmarks.

The EDM parameters were experimentally optimized through trial runs (some from literature as shown in Table 1) to ensure stable machining, controlled hardness, and effective material removal for H13 steel dies.

The dies were manufactured using H13 die steel, with its chemical composition presented in Table 2. The elemental composition was verified through spectrographic analysis.

Table 2: Material Composition of Die [16]

AISI H13	Ni	Mn	C	Mo	Si	V	Cr	Fe
wt%	0.19	0.28	0.48	0.93	1.04	1.12	5.39	Rest

The threading dies were prepared with the dimensions listed in Table 3. Fabrication was carried out using both conventional machining processes (turning, drilling, and tapping) and non-conventional methods (EDM wire cutting and EDM die sinking).

Table 3: Die Dimensions

Die Segments	Dimensions
Outer Diameter	38 mm
Inner Diameter	10 mm
Thickness	13 mm
Side Holes Diameter	10 mm
Type of Material	H13 Die Steel

In conventional machines, the direct interaction between the cutting tool and the workpiece induces substantial alterations in the material's internal structure, which can lead to early die failure. Conversely, non-conventional machining, such as EDM, eliminates mechanical contact, thereby preserving structural integrity. To comparatively assess the effects of both machining approaches, experiments were conducted under equivalent conditions, with parameters standardized based on material removal rate. The specific parameters used for conventional machining are presented in Table 4.

Table 4: Machining Parameters

Speed	Feed	Depth of Cut	MRR
310	0.06	0.2	480.4

The machining framework for the non-conventional machining process, as presented in Table 4, is structured with reference to the material removal rate (MRR). Two identical threading dies, fabricated from the same material, were employed for external threading operations to facilitate the analysis of machining effects.

3.1 Analysis Parameters

- I. Microstructure analysis
- II. Effect on Hardness
- III. Die Strength (No of Pieces Produce)

The primary objective of this study is to compare the technical advantages of non-conventional machining over conventional methods in die fabrication. Non-conventional processes, capable of shaping materials from simple to highly complex geometries, were employed to minimize the adverse effects of machining on dying integrity. Cracks observed during production were attributed to material distortion caused by machining stress. The optimal machining process was determined based on three key criteria: higher number of parts produced, lower post-machining hardness, and larger grain area observed in microstructural analysis ensuring enhanced die performance and durability.

3.2 Experimental Methodology and Process Design

Experimental Methodology and Process Design are as follows:

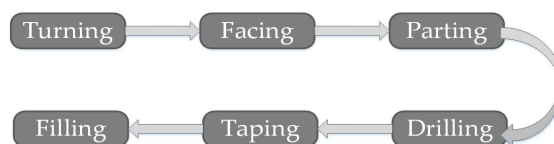


Figure 1 process plan for conventional die

The conventional die fabrication sequence, illustrated in Figure 1, involves turning, facing, and parting, followed by drilling, tapping, and filling. An iterative loop from parting to drilling emphasizes precise dimension control before threading and final finishing operations.

3.2.1 procedure

A die steel workpiece measuring 60 mm in length and 40 mm in width was selected for conventional machining. The surface was turned and faced, followed by cutting into 14 mm segments according to the specified die dimensions. Drilling operations were then performed to create 10 mm diameter holes. Tapping was carried out using a sequence of taper, second (bottoming), and plug taps. After side hole tapping on the die face, the finishing process was completed through manual filing. The final holes maintained a consistent diameter of 10 mm, as illustrated in Figure 2.



Figure 2 conventional machining operation

3.2.2 EDM machining

The fabrication of the non-conventional die using EDM machining was initiated in accordance with a structured process plan, following the sequential stages typically observed in conventional die manufacturing. The process begins with parting and tool preparation, followed by center hole drilling, tapping, side hole machining, and final filling. An iterative loop returning to the tapping stage after center hole formation is incorporated, as illustrated in Figure 3.

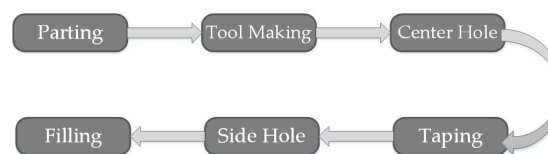


Figure 3 process plan for Non-conventional Die

3.2.3 Procedure

In this experiment, EDM metal cutting and EDM mold opening processes were employed for the fabrication of the non-conventional die. Copper tools were utilized as electrodes during the EDM mold opening, with copper rods selected to match the required final dimensions of the mold cavity. EDM metal cutting was applied for precise separation tasks, ensuring minimal impact on the workpiece. Centering operations were also performed using EDM, as illustrated in Figure 4. The copper electrode diameter was equivalent to the target hole size, allowing for accurate cavity formation. Threading was subsequently performed using taps, followed by EDM machining to complete and refine the side holes.



Figure 4 non-conventional machining process

3.2.4 Hardness testing

Following the machining operations, hardness testing was conducted on both dies to evaluate the impact of the respective machining processes on material properties. The parameters used for hardness testing are presented in Table 5, while the results are detailed in the Results.

Table 5: Rockwell Hardness Test Parameters

Hardness Test	Parameters
Load	150 kg
Indenter size	120° diamond Sphero conical
Indenter	Diamond
Machine	Rock well Hardness Testing
Unit	HRC
Standard	ASTM E18

Following the hardness testing, the microstructure of the dies was analysed to assess the effects of machining. Grain size determination was carried out in accordance with ASTM E112 using the planimetric method. Detailed microstructural results are presented in the Results section. Subsequently, a heat treatment process was performed to enhance the dies' capability for threading mild steel rods, enabling evaluation of die life and performance.

3.2.5 Heating

Electrical Furnace is used (as shown in figure 5) for Heat Treatment Process of Threading Dies [17], which has temperature range approximately 1000 Celsius and then we use the water for the quenching of Dies for hardening purpose. Both Conventional and non-Conventional tools can be heated in the oven. Preheat the mold to 800 degrees Celsius and then let the mold sit for 40 minutes (1 inch steel requires 1 hour of holding time). The heating temperature of the two Molds (normal mold and unusual mold) is the same. The following is the heating temperature of the furnace.

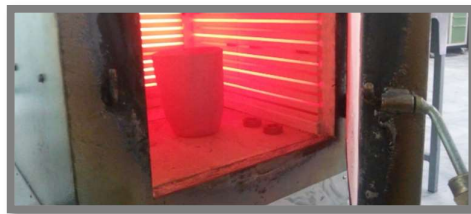


Figure 5 Heat treatment Dies

3.2.6 Quenching

After the mold is completed, water is used as a quenching medium to increase the hardness of the mold. After quenching the hardness of dies are tested. As the hardness due to quenching increase from the range of hardness which is defined for dies. Then tempering is needed for to achieve require level.

3.2.7 Tempering of Threading Dies

Higher temperatures result in higher ductility and lower quality and hardness [18]. Low treating temperature does not provide ductility but provides high strength and hardness. For mold tempering, we will preheat the mold at 500 degrees Celsius and give it a holding time of about an hour before removing the mold from the oven and cooling it at room temperature to keep our conditions bad.

4. Results and Discussion

Analysis of this research is based on three factors, and every factor has its own importance. The following are the parameters of this research.

4.1 Hardness

Table 6 shows that the hardness of Conventional Die increased during the machining, and it is the cause of failure. Increase in the hardness after machine is dangerous because when we heat-treated a hard thing it adopts crack in its internal structure, so this may cause of failure of dies at the initial stage.

Table 6: Hardness Comparison

Dies	Hardness (HRC)			
	Original Material	After Machining	After Heat Treatment	After Temperature (Tempering)
Conventional	18	28	52	44
Non- Conventional	18	19	48	42

The observed changes in the internal structure of the material are attributed to elevated cutting temperatures, mechanical stresses, and direct tool–workpiece contact inherent in conventional machining. These factors result in increased hardness, necessitating post-machining heat treatment to achieve the desired hardness range of 40–45 HRC for threading dies. Following water quenching, die hardness increased to 48–52 HRC, exceeding the required range; therefore, tempering was performed to reduce hardness. After tempering, the hardness of both dies was adjusted to 42–44 HRC, aligning with operational requirements. Since the non-conventional (EDM) process exhibited negligible hardness variation post-machining, it was deemed more suitable for die manufacturing. However, due to the relatively lower hardness of EDM-fabricated dies, this method may be less appropriate for applications requiring high-strength mold performance.

4.2 Microstructure Analysis

Planimetry involves precisely counting the number of objects in a known region. The number of grains per unit range NA is utilized to decide the ASTM grain measure number G. The precision of this strategy may be a work of grain number.

4.2.1 Non-Conventional Grain Size

Microstructure research reveals that the original picture displays the spacing between the grains and the prepared picture appears the grain number for Non-Conventional Pass on is 12.466 figure 6 shows the result of microstructure test.

Grain Number	12.466	Result	5.6µm
Preparation	MICRO	Calibration	xel= 0.00012 Millimeters
Specimen	Un-Conventional Die	Scanned Area	0.021020318
Original Image		Processed Image	
			

Figure 6: Non-Conventional Grain Size results

The data presented in the table 7 indicate that the average grain diameter is approximately 5.6 µm, with a corresponding average grain area of 31.5 µm².

4.2.2 Conventional Die Grain Size

Microstructural analysis of the conventional die revealed distinct grain boundaries in the initial image, while the processed image yielded a grain number of 13.58. The corresponding grain diameter and structure are illustrated in Figure 7.

Grain Number	13.581	Result	3.3 μ m
Preparation	MICRO	Calibration	xel= 0.00012 Millimeters
Specimen	Conventional Die	Scanned Area	0.021020318
Original Image		Processed Image	
			

Figure 7 Conventional Die Grain Size results.

According to the standard grain size reference table, the average grain diameter for the conventional die is 3.3 μ m, with an average grain area of 11.1 μ m². A comparative analysis of these results is presented in Table 7.

Table 7: Comparison of the results.

	Grain Number	Average Grain Diameter (μm)	Average Grain Area (μm²)
Conventional Die	13.581	3.3	11.1
Non-conventional Die	12.46	5.6	31.5

4.2.3 No of Parts to be produced

The third parameter of this research is part to be produced. The following are the results of this parameter.

Conventional Die: Dies are used to cut threads on small steel bars. We form threads in the middle of the stick. The tool is made for MS bar thread lengths up to 16 inches (406.6 mm). It breaks and its efficiency is low due to the Hardness increase and change its internal structure.

Non-Conventional Die: These dies are very strong and very efficient these are also used for the purpose of threading on the surface of MS steel rod. Non-conventional Dies make thread up to 20 inches (508 mm) and these dies are now in working.

Following are the final dies both Conventional and Non-Conventional with parts produced by both types of dies as shown in figure 8.



Figure 7: Number of Parts to be produced.

5 CONCLUSION & RECOMMENDATIONS

5.1 Conclusion

So, as a result, this study highlights the significant differences between conventional and non-conventional machining techniques in die fabrication, particularly in terms of

material properties, efficiency, and failure mechanisms. The experimental results demonstrate that conventional machined dies are more prone to cracking, as evidenced by the substantial increase in hardness, the study demonstrated a significant difference in performance during threading operations. Conventional dies failed after threading up to 16 inches, whereas non-conventional dies were able to successfully produce threads up to 20 inches, indicating a longer operational lifespan. The observed mechanical stresses induced by conventional machines, due to direct tool-workpiece contact, contributed to increased hardness and were a primary factor leading to early die failure. Overall, the results underscore the superior advantages of non-conventional machining techniques, such as EDM, in die manufacturing, including improved precision, reduced hardness variations, and enhanced durability, which collectively offer a more reliable and efficient solution for die fabrication.

5.2 Recommendations

In future this research work can be applied on a broad level in the industries for the manufacturing process of different dies.

- 1) Other tool steels or advanced materials with superior machinability, wear resistance, and thermal stability may be considered to enhance productivity. Potential alternatives include:

AISI D2 Steel High wear resistance and good toughness.

AISI P20 Steel Suitable for plastic mold applications with better machinability.

AISI M2 High-Speed Steel Offers excellent hardness and heat resistance.

Titanium Alloys (Ti-6Al-4V) Lightweight with high strength and corrosion resistance.

Carbide Composites Enhanced durability for high-performance applications.

- 2) Advanced machining techniques like **Water Jet Machining (WJM)**, **Electrochemical Machining (ECM)**, and **other precision manufacturing methods** may be used to improve efficiency, accuracy, and overall part quality. The selection of the most suitable process should align with the material properties and desired performance characteristics.

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